

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 20

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Copying the structural / functional adaptations of living organisms in the design of manufactured items (1) • Aircraft wingtip fins copied from splayed wingtip feathers of soaring birds, reducing turbulence and fuel use (1) • Humpback whale flipper tubercles copied on ship rudders and wind turbine blades (tighter turns, rotation at lower wind speeds) (1) • Shark skin ridges reducing friction / bacterial adhesion (surface coatings, hospital surfaces) (1) • Termite mound convection copied for natural ventilation of buildings; hollow strutted bird bones in lightweight roofs; gecko toe adhesion; burr hooks - Velcro; lotus leaf self-cleaning glass; spider silk for airbags / body armour; bat / dolphin echolocation - ultrasound scanners (1)	4
01.2	<i>1 mark for each point.</i> • Plants produce toxic chemicals (eg alkaloids) to defend themselves against herbivores; in controlled doses these affect human physiology (1) • Examples: morphine / codeine from poppies; quinine from cinchona bark (malaria); aspirin from willow bark; taxol from yew (cancers); diosgenin from Mexican yam (contraceptives, cortisone) (1) • Animal examples accepted: AZT from a marine sponge (HIV), shark liver lipids (tumours) (1)	3
01.3	<i>1 mark for each point.</i> • Only a very small proportion of species have been studied for medicinal substances so unknown species may contain valuable chemicals (1) • Conserving the habitat protects all its species, including those not yet investigated (and their interdependent relationships) (1)	2

Question	Marking guidance	Marks
01.4	<i>1 mark.</i> - Squid giant nerve cells - nerve function / heart disease, stroke; - marsupials (kangaroo, wallaby pouch) - embryo development; hippo skin secretion - sunscreen / burns; sea urchin embryos - teratogen testing; armadillo - leprosy; marine sponge proteins - transplant rejection (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-15; AQA spec 3.1.2.1.1-3.1.2.1.2.	
02.1	<i>Up to 6 marks in total; award marks in proportion to the completeness of each point.</i> - Non-indigenous species (Sitka spruce from North America) chosen for fast growth and pest resistance; indigenous wildlife's needs for food, pollination, seed dispersal and habitat are not met (up to 2) - Single species: easy planting, thinning and felling over large areas; but each tree species supports its own community so few species overall (up to 2) - Close planting makes trees compete for light so they grow tall and straight (good timber) but shades out ground flora and the species that depend on them (up to 2) - Simple age structure: uniform size makes management easy but gives little variety of light, temperature, wind; no old trees or dead wood for invertebrates and woodpeckers (up to 2) - Mixed woodland: varied niches, stratification, dead wood, indigenous trees each with their own dependent species (up to 2)	6
02.2	<i>1 mark for each point.</i> - Trees are felled when income will be greatest - approaching full size when the growth rate starts to decline (1) - Continuing to grow them ties up land and capital for little extra timber; natural forests keep old and dead trees (1)	2
02.3	<i>1 mark for each point.</i> - Industrial society replaced species-specific uses (baskets, fencing, charcoal, tool handles) with large-scale uses: chipboard, plywood, pulp (1) - Wood fuel replaced by fossil fuels / electricity; so traditional management declined and woods became uniform (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 365-367; AQA spec 3.5.3.2.	
03.1	<i>1 mark for each point.</i> - No major temperature extremes; regular water supply with no pronounced dry season; deep fertile soils built by leaf litter and root binding; strong seasonality (1) - Deciduous canopy present spring to autumn shades the floor during the best growing temperatures (1) - Bluebells, snowdrops, lesser celandine grow and flower early in spring before the trees leaf (1) - Winter food shortage causes animals to hibernate (dormice), store food or migrate (1)	4

Question	Marking guidance	Marks
03.2	<i>1 mark for each point.</i> • Long time for species to colonise; some woodland plants colonise very slowly so their presence (ancient woodland indicators) shows the wood is old (1) • Natural woodland has trees of all ages with clearings where old trees died, giving varied light and dead wood habitats (1) • A new plantation has a single-age dense canopy with low light and few niches until trees start to die (over 200 years) (1)	3
03.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Clearance for farmland, plantations, urban development, roads, mineral extraction (1) • Fragmentation isolating remaining woods so populations cannot move between them and are vulnerable (1) • Management change: end of coppicing so woods become uniform and dark; replacement by conifer monocultures (1) • Overgrazing by deer preventing regeneration; introduced pathogens (ash dieback) (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	
04.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • EDGE: Evolutionarily Distinct and Globally Endangered - few close relatives, often the last species of their genus / family (Bactrian camel, pygmy hippo, northern bald ibis); loss would remove a unique branch of evolution / genes (2) • Endemic: found in only one area (often islands - Galapagos, Madagascar, Seychelles: Aldabra giant tortoise, red ruffed lemur); if lost locally there is no population elsewhere to recolonise (2) • Keystone: disproportionately important for the structure of the community relative to its abundance - predation of dominant species, seed dispersal, habitat creation (jaguar, forest elephants keeping paths and water holes open, wolves, beavers); losing them causes loss of many other species (2) • Flagship: high public profile (tiger, panda, elephant, orang-utan) raising funds and support that also protects the whole habitat and less charismatic species (fungi, bats, moths) (2)	8
04.2	<i>1 mark for each point.</i> • A fragmented population may consist of isolated groups each too small to be viable (resources, inbreeding) even if total numbers are large (1) • A species confined to a single area (eg bonobos in one country) is vulnerable to any local change / conflict, unlike chimpanzees spread across many countries (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 28-29; AQA spec 3.1.2.3.1.	

Question	Marking guidance	Marks
05.1	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Success: Cactoblastis moth introduced from South America controlled prickly pear cactus in Australia; Encarsia parasitic wasps control greenhouse whitefly (up to 2) • Problems: cane toads introduced to Australia to control sugar cane beetles did not control them but ate many other species and are toxic to predators; rosy wolfsnail introduced to Polynesia to control African land snails ate endemic Partulid snails (48 species extinct); harlequin ladybirds introduced to Europe eat native ladybirds (up to 2) • Lesson: control agents should be specialist feeders on the pest (up to 2)	4
05.2	<i>1 mark for each point.</i> • Release an artificial sex scent attracting males (or females) of the pest species (1) • Used to monitor when the pest arrives so pesticide is timed correctly, or (in enclosed greenhouses) to trap enough of one sex to prevent breeding (1)	2
05.3	<i>1 mark for each point.</i> • Regular low doses kill only the most sensitive bacteria so resistant strains survive and multiply (1) • Agricultural use exceeds medical use, so resistance spreads (1) • Resistant zoonotic bacteria (E. coli, Salmonella, Campylobacter) can transfer to humans and are harder to treat; hence reduced use of antibiotics is a sustainability goal (1)	3
05.4	<i>1 mark.</i> • Grow nectar-rich flowering plants alongside crops; restrict pesticides harmful to bees / spray when not in flower; introduce bee hives (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 316-320; AQA spec 3.5.1.1.	
06.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Carnivores fed pellets of sandeels, herring, anchovy; overfishing sandeels reduces puffins and terns; reduced by more plant protein / vegetable oil in pellets or farming herbivores (tilapia) (2) • High density spreads lice (which spread to wild fish) and infections; pesticides (organophosphates, pyrethroids, hydrogen peroxide) kill wildlife; antibiotics breed resistant bacteria; reduced by lower densities, single-age tanks with flow from young to old, wrasse as biological control, warm-water washes, circulating flow (2) • Escaped farmed fish breed with wild fish introducing disadvantageous genes (bright colour), or non-indigenous escapees colonise as predators / competitors; reduced by better cage design, sterile triploid fish (pressure-treated eggs), species that cannot breed locally (Japanese oysters in UK) (2) • Faeces and uneaten food decompose causing deoxygenation under cages; reduced by monitored feeding, siting cages in strong currents, effluent treatment / reed beds for freshwater farms, IMTA with seaweed and shellfish absorbing wastes (2)	8

Question	Marking guidance	Marks
06.2	<i>1 mark for each point.</i> • Predators are killed under licence to protect stock, reducing wild populations of protected / valued species (1) • Better cage designs, netting, fencing, bird scarers exclude predators without killing (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 356-359; AQA spec 3.5.2.3.	
07.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $600 + (90 + 150) - (220 + 80) = 540$ (1) • Stock fell by 60 so the catch exceeds the MSY / unsustainable (1)	2
07.2	<i>1 mark for the correct answer.</i> • $A + G - M = 90 + 150 - 80 = 160$ (1)	1
07.3	<i>1 mark for each point.</i> • Catch is landed and weighed at ports so it is recorded (1) • Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed (1) • Representative sampling of a large enough proportion of the area is impractical / expensive; survival of eggs and larvae varies greatly between years (1)	3
07.4	<i>2 marks for each point (partial credit for a partly correct point).</i> • CPUE: if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained; a better measure than catch size which technology can prop up (2) • Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish (2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2.	
08.1	<i>1 mark for each point.</i> • Hydrosere (1) • Single-celled algae (and floating plants) - arrive on birds / insects / from soil (1)	2
08.2	<i>1 mark for each point.</i> • Pioneers and later plants grow and die; dead organic matter plus sediment washed in fill the pond so water becomes shallower (1) • As depth falls rooted plants (pondweeds then emergent reeds with leaves above water) can anchor and colonise; emergents shade submerged plants which die out (1) • Sediment / soil accumulates and becomes able to support larger plants; water-tolerant trees (willow, alder) colonise the waterlogged soil (1) • Transpiration by trees and further sediment dry the soil and make it aerobic so oak can colonise; the denser canopy shades out smaller plants (1) • Each community changes abiotic conditions (depth, light, nutrients, oxygen), making it suitable for the next colonisers which out-compete them; conditions become less extreme and biotic factors increasingly important (1)	5

Question	Marking guidance	Marks
08.3	<i>1 mark for each point.</i> - Starting conditions (bare rock vs open water) differ but each stage makes conditions less extreme and builds soil / organic matter (1) - The final community is controlled by the regional climate (temperature, water availability), which is the same for both - climatic climax (eg oak woodland in the UK) (1)	2
08.4	<i>1 mark.</i> Lichens and algae colonise bare rock (extreme temperatures, no soil, little water); rock fragments and dead organic matter accumulate; mosses, then grasses and ferns as thin soil forms; wind-dispersed trees (birch) then animal-dispersed (oak, beech) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 73-76; AQA spec 3.1.3.3.	
09.1	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Hydrological cycle (1) - Carbon cycle (1) - Nitrogen cycle (1) - Atmospheric circulation (1) - Thermohaline circulation (1) - Photosynthesis (1)	2
09.2	<i>1 mark for each point.</i> - Enzymes reduce the activation energy of reactions (1) - So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures (1)	2
09.3	<i>Any 1 from the points below - 1 mark each, maximum 1.</i> - Haber process (nitrogen fixation) (1) - Incineration of wastes (decomposition) (1) - Many manufacturing processes (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • In-situ: species remain in natural conditions with the whole community and inter-species relationships; protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers) • Limits of in-situ: external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations • Ex-situ: captive breeding when in-situ is impossible or insufficient; selection criteria; difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation); techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning); seed banks; successes (Hawaiian goose, Iberian lynx, red kite, large blue) • Limits of ex-situ: expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring) • Judgement: ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to; both needed, in-situ fundamental	9
10.2	1 mark. • Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Resources: wood, fibres, oils, fuels, food; new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • Biomimetics: wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • Medicines: poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks; few species yet studied; physiological research (marsupials, hippo, sponges) • Genetic resources: crop wild relatives, Vavilov centres, gene pools and inbreeding • Ecosystem services: atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • Evaluation: economic vs intrinsic / moral arguments; services unnoticed because of dynamic equilibrium; unknown future value; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	25

Question	Marking guidance	Marks
11.2	Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Ecological features: cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web; specific abiotic requirements (temperature, light, salinity, low turbidity) • Importance: quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • Vulnerability: narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching; slow growth so slow recovery; physical damage; overfishing and food-web effects (crown of thorns); introduced species; dependence on mangroves and seagrass • Conservation: MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	25